6. Vehicle Service Management Automation Suite & Workflow Options

Aim

To configure Vehicle Service Management automation in Zoho Creator using form, schedule, approval, blueprint, batch, report, and function workflows.

1. Objective

This experiment demonstrates eight Zoho Creator workflow options for the vehicle service lifecycle, from request submission through inspection, repair approval, billing, and pickup.

2. Workflow Mapping Overview

Workflow Type	Configuration / Purpose
Form Workflow	On_Service_Request_Submit_Workflow – triggers acknowledgement emails and service advisor notifications upon service request submission.
Schedule	Daily_Service_Followup_Reminder_Schedule – automates daily reminders for pending parts or vehicle verification reminders.
Approval	HighCost_Repair_Approval – manages multi-level service manager approval for high-cost repairs.
Payment	Service_Bill_Payment_Workflow – handles the automated service bill payment process.

Blueprint	Vehicle_Service_Lifecycle_Blueprint – establishes a controlled vehicle service lifecycle with defined stages and transitions.
Batch Workflow	Bulk_Service_Reminder_Batch – runs nightly batch reconciliations for service-history updates.
Report Workflow	OnEdit_Recalculate_Service_Bill_Workflow – recalculates service eligibility and final-bill metrics after report-level edits.
Function	Generate_Service_Invoice_Function – reusable Deluge function for automated service invoice communications.

3. Detailed Workflow Configurations & Evidence

Form Workflow (On_Service_Request_Submit_Workflow)

- **Purpose:** Automatically triggers an email notification to the customer and internal service operations team immediately upon successful creation of a record in the Vehicle_Service_Request_Form.
- **Action:** Configured with an acknowledgement email action (Send mail to zoho.adminuserid) executed without conditions upon form creation.

Schedule Workflow (Daily_Service_Followup_Reminder_Schedule)

- **Purpose: Runs daily at 10:00 AM to check for pending service follow-ups (such as vehicle registration and service-history documents).**
- **Action: Automatically triggers reminder notifications to customers or service advisors regarding incomplete inspections or service records.**

Approval Workflow (HighCost_Repair_Approval)

- **Purpose: Enforces structured repair approval levels for high-cost repair requests submitted via the Vehicle_Service_Request_Form.**
- **Events:** Manages "On Start", "Level 1 Approver" actions, along with distinct branching logic for Approve (sending repair-approval confirmation messages to customers) and Reject outcomes.

Blueprint (Vehicle_Service_Lifecycle_Blueprint)

- **Purpose: Governs the progression of service requests through defined lifecycle stages.**
- **Stages: Requested, Inspection, Approved, and Ready for Pickup.**

Batch Workflow (Bulk_Service_Reminder_Batch)

- **Purpose: Executes automated vehicle service-history updates across blocks of records (e.g., in batches of 1,000 records) during off-peak hours.**
- **Deluge Integration: Flags service records for manager review when limits are exceeded.**

Function Implementation (Generate_Service_Invoice_Function)

- **Purpose: Fetches a service record by ID, validates it, and emails an invoice using Customer_Name, Customer_Email, and Estimated_Service_Cost.**

Screenshots:

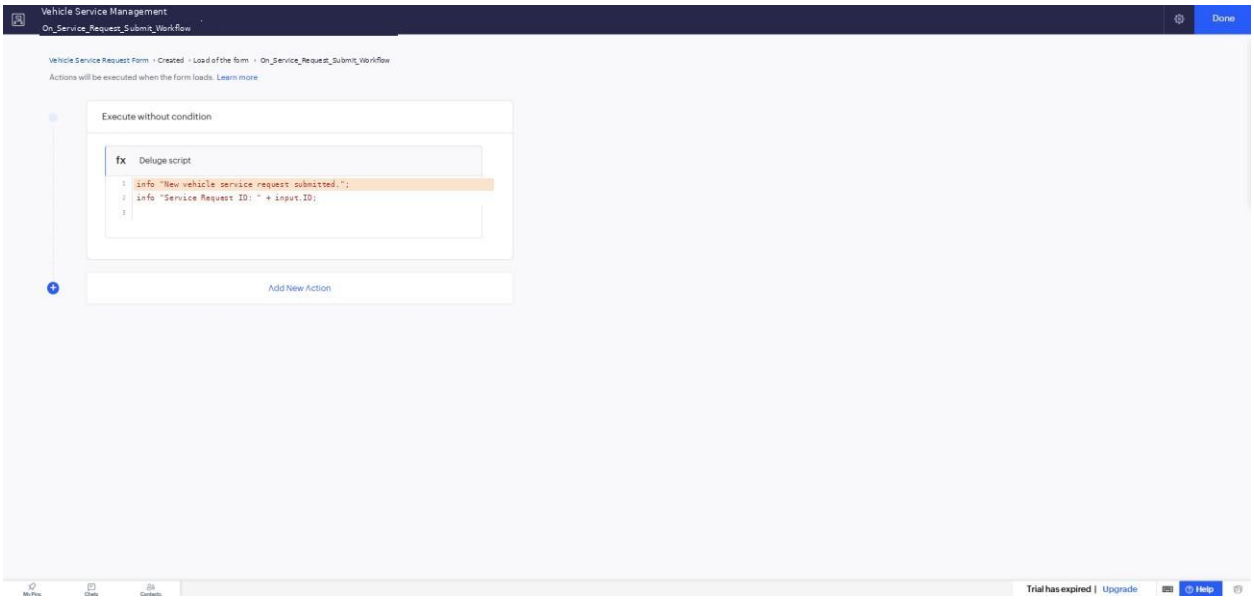


Figure 1 – Form workflow action: acknowledgement email configured for successful submission

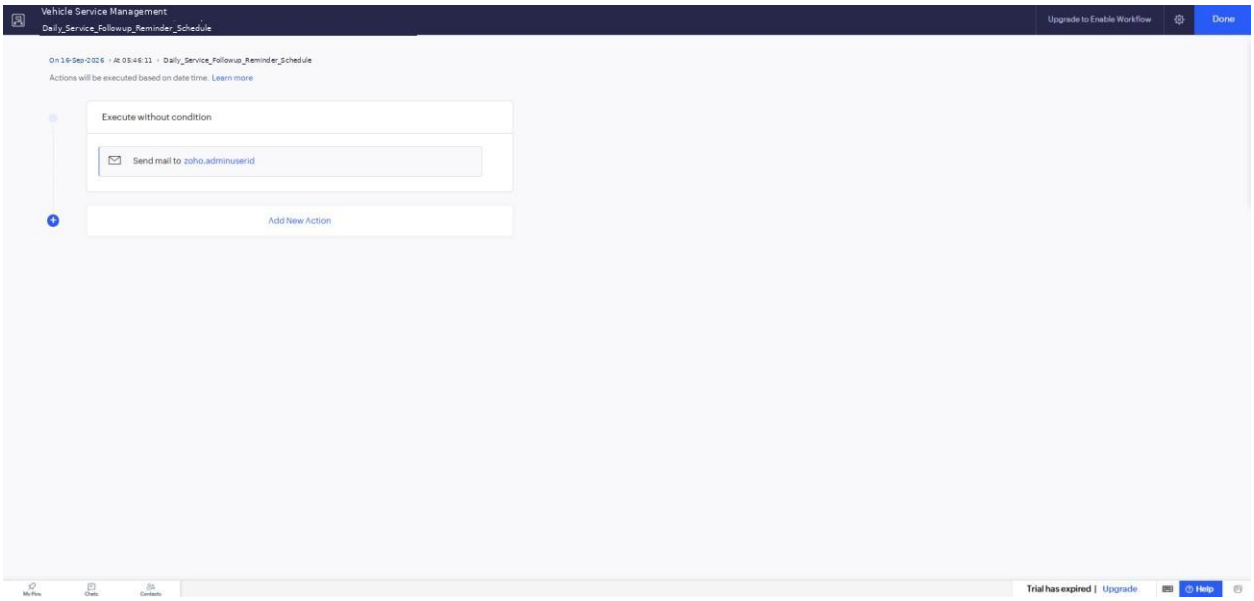


Figure 2 – Schedule workflow: Daily_Service_Followup_Reminder_Schedule configuration.

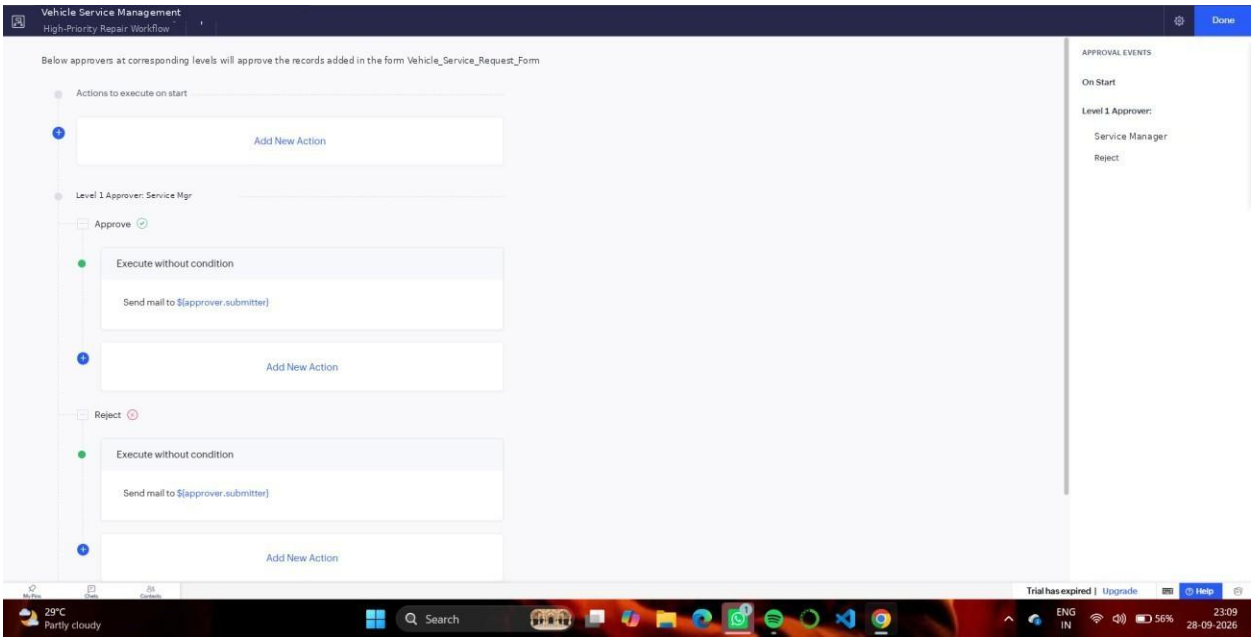


Figure 3 – Approval workflow: HighCost_Repair_Approval with approve/reject events.



Figure 4 – Blueprint: Vehicle_Service_Lifecycle_Blueprint showing stages and transitions.

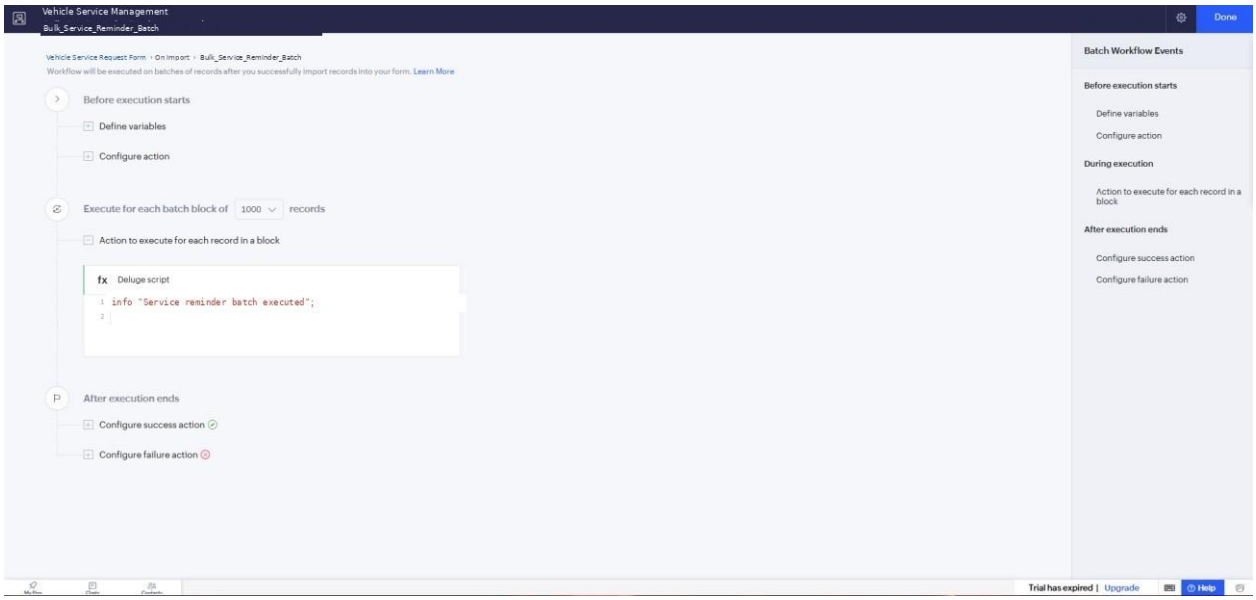


Figure 5 – Batch workflow: Bulk_Service_Reminder_Batch with batch execution and Deluge action.

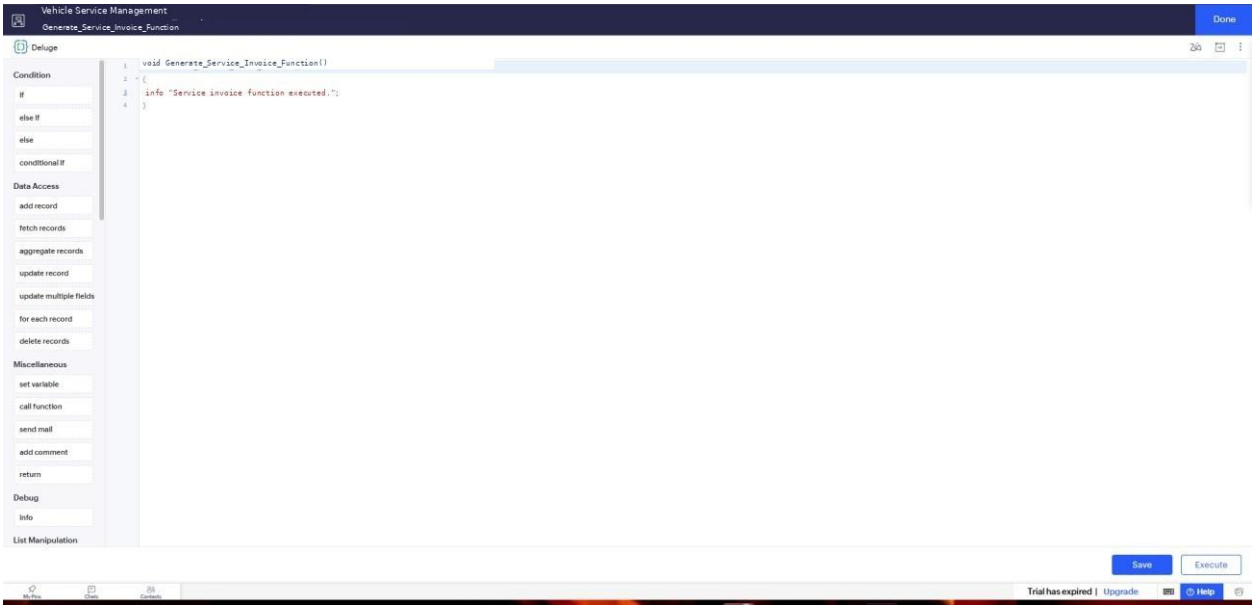


Figure 6 – Function: Generate_Service_Invoice_Function with Deluge implementation

Result

The **Vehicle Service Automation Suite** was successfully designed, configured, and documented using Zoho Creator's advanced workflow builder framework, covering form triggers, schedules, approvals, blueprints, batch processing, and custom Deluge functions tailored specifically to vehicle service operations.